Problem 4: Can a variational autoencoder over-fit?

Yes, a variational autoencoder can overfit.
Although VAEs are often considered more robust than standard autoencoders due to the regularization induced by the KL divergence term, they can still overfit when the model capacity is high or the dataset is small.
Overfitting in a VAE typically occurs when the encoder and decoder networks become powerful enough to effectively memorize the training data. In such cases, the latent representations no longer follow the intended prior distribution, and instead encode highly specific information about individual training samples.
One key mechanism leading to overfitting is when the KL divergence term becomes too weak relative to the reconstruction loss. In this regime, the model behaves similarly to a standard autoencoder, focusing primarily on minimizing reconstruction error rather than learning a meaningful latent distribution. As a result:
①Reconstruction quality on the training set becomes very high
②Generated samples (from the prior) are poor or unrealistic
③The latent space loses smoothness and structure
In summary, while the KL term provides regularization, it does not completely prevent overfitting, especially when the model is large or improperly balanced.

Problem 5: Can a diffusion generative model over-fit?

Yes.
Diffusion models are trained to learn the reverse process of a stochastic noise-adding procedure, typically by predicting noise or estimating the score function. Because of this, they do not directly memorize individual data points in the same way as deterministic models.
However, overfitting can still occur when:
①The dataset is small
②The model capacity is large
③Training is prolonged without sufficient regularization
In such cases, the model may learn a very accurate approximation of the training data distribution, but with limited generalization ability. Instead of capturing the true underlying distribution, it effectively memorizes the training data at the distribution level.
This leads to:
①Reduced diversity in generated samples
②Generated outputs that resemble training examples too closely
③Poor generalization to unseen variations

Problem 6: Latent Space Interpolation
(a) Why might a geodesic path on a sphere lead to better results than a linear path?

In a VAE, the latent space is typically regularized to follow a multivariate standard normal distribution. In high-dimensional spaces, samples from a Gaussian distribution tend to concentrate on a thin spherical shell rather than near the origin.

When performing linear interpolation between two latent vectors, the path often passes through the interior of this sphere. These regions correspond to low-probability areas under the prior distribution, meaning that the decoder is less well-trained on such inputs. As a result, decoding these intermediate points can produce unrealistic or distorted outputs.
In contrast, a geodesic path on the sphere stays on the high-probability region of the latent space. Since the model has been trained on samples drawn from this region, the decoder is more likely to produce meaningful and smooth interpolations.
Therefore, geodesic interpolation better respects the geometry of the latent distribution, leading to higher-quality image resynthesis.

(b) How would you choose the best geodesic path on the sphere?

① Choose the path that maximizes the likelihood of decoded images under the model. This ensures that intermediate points correspond to realistic samples.
②Evaluate how well decoded images along the path preserve meaningful structure. A good path should produce smooth and semantically consistent transitions.
③Use data-driven methods (e.g., principal component analysis or latent direction discovery) to identify meaningful directions in the latent space. The geodesic path can then be aligned with these directions to ensure interpretable changes (e.g., rotation, lighting, object attributes).
④ Select the geodesic that minimizes abrupt changes in decoded outputs, ensuring smooth transitions between the endpoints.